

Proposal for IP block development for the opencircuitdesign 2026 Chipalooza challenge

IP Block name:

Single ended to differential and differential to single ended analog pin buffers

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Circuit description

On-chip analog design building blocks (amplifiers, filters, comparators, ...) should be fully differential in/differential out, for signal range and PSRR.

Multi-user shared design platform ASICs like Chipalooza, TinyTapeout, HeiChips, etc. are severely I/O limited.

To alleviate the scarcity of I/O pins while facilitating fully differential designs in slots assigned to users, some building blocks are useful:

- differential-in to single-ended out class AB analog pin driver
- single-ended analog pin to fully differential internal signal buffer amplifier

Generally, I'm more interested in all sorts of building blocks useful for an on-chip potbox [<http://opencircuitdesign.com/~tim/research/potbox/potbox.ps>] than a self-contained circuit presentable as a student project.

To still allow a self-contained, demonstrable proof-of-concept slot, I intend to add

- an internal common mode voltage reference to provide a sufficiently stable V_{CM} for internal common mode feedback circuits
- a proof-of-concept fully differential circuit (e.g., a $g_m C$ quadrature oscillator with differential input)
- (if necessary) a buffered power supply for input and output pin buffers for PSRR

Circuit pinout

from infrastructure

Pin name	Use	Value
vdd3v3	analog power	3.3V nominal
vdd1v2	digital power	1.2V nominal
vss3v3	analog ground	0V
vss1v2	digital ground	0V
vssio	I/O ground	0V
ibias	bias current	to be discussed, 100nA ballpark
igmc	$g_m C$ control current	to be discussed
vbias	common mode voltage	to be discussed, if adjustable

dedicated analog pins (maybe pins with integrated buffer circuits)

Pin name	Use	Value
vin	single ended analog in	must include vssio, should include vdd3v3
vout	single ended analog out	

shared analog pins

Pin name	Use	Value
vcm	common mode voltage	maybe switchable to internal V_{CM}
vdiffp	differential analog I/O	"internal" differential voltage (buffered?)
vdiffrn	differential analog I/O	"internal" differential voltage (buffered?)

shared digital input pins

Pin name	Use	Value
ena	master enable	
reset	reset digital logic	reset scan chain (and other logic?)
outbuf_en	output pin buffer enable	
inbuf_en	input pin buffer enable	input of next internal stage to V_{CM} if off
filter_en	internal g_mC filter enable	inbuf connected to outbuf if disabled
vdif_en[1:0]	differential analog I/O enable	off, in, out, out_offset
vcmsel[1:0]	V_{CM} select	analog pin, internal, infrastructure
scanclk	scan chain shift clock	
scanctrl[1:0]	scan chain control	idle, shift, write, readback
scanin	scan chain in	

shared digital output pins

Pin name	Use	Value
scanout	scan chain out	

Note any changes from the specification, such as if trim bits have been added.

Circuit architecture

- differential input class AB pin driver (integrated in pin?)
- single ended input pin to differential signal input buffer (unity+negative unity gain, integrated in pin?)
- switch logic to select common mode voltage (infrastructure, shared analog pin, or internal circuit)
- switch logic to select bias current (infrastructure or internal, see <https://github.com/tatzelbrumm/ihp-sq13cmos5l-librelane-template/tree/tatzelbranch/xschem>)
- fully differential g_mC filter, control current provided by infrastructure or a variant of [the current DAC taped out to HeiChips](#)
- a [digital scan chain comprising a shift register and holding registers](#) for digital control as necessary

External resources (if any) (all resources must be open source)

List any resources that come from on chip, such as a bandgap-referenced voltage or PTAT current, or from other IP blocks (such as a comparator used in an ADC). List any external resources needed for testbench circuits, including any digital control sequencing needed.

If possible, I will derive the pmos and nmos bias currents from layout structures that match the layout of the current and voltage references on the Chipalooza harness.

Maybe I will cheat and design yet another of my favorite [Oquey&Aebischer bias](#) blocks [\[https://www.semanticscholar.org/paper/Ultra-High-Input-Impedance%2C-Low-Noise-Integrated-Chi-Maier/ab83669efb5f29a94e19b8e2c3f4801ab50ba3ea/figure/10\]](https://www.semanticscholar.org/paper/Ultra-High-Input-Impedance%2C-Low-Noise-Integrated-Chi-Maier/ab83669efb5f29a94e19b8e2c3f4801ab50ba3ea/figure/10), [\[https://github.com/MastellaM/sky130_TAC3/pull/3\]](https://github.com/MastellaM/sky130_TAC3/pull/3)

with matching layout primitives.

Specification difficulties

List all specifications which may be difficult to attain, and what circuit design methods will be used to meet those specifications. Note where specifications will be affected by layout considerations, such as mismatch, crosstalk, and I-R drop.

Bias currents, bandwidth, SNR and PSRR, Class AB output current and load capacitance are to be determined from what other participants in Chipalooza may find useful **(to be discussed)**. By default, I'd aim for low noise, low (10...100nA branch) current for internal amplifiers and some 10mA class AB output current into 100pF.

Unless advised otherwise, I would aim for low (<1 MHz) bandwidth. If the input and output pin buffers become more useful to others, I'll aim for higher bandwidth.

Claude tells me:

Calibration tiers, output buffer (differential in → single-ended pin):

	GBW	FPBW @1 V _{pk}	Risk
Conservative	1–5 MHz	~1–2 MHz	first silicon works, no drama
Reasonable	10–30 MHz	5–10 MHz	the sweet spot; poles comfortably clear
Aggressive	50–100 MHz	20 MHz	pole management is now the whole design; MOScap C _c uncertainty becomes acute

Don't >200 MHz —stop calling it a buffer

Testbenches required for verifying circuit performance

List what testbenches are used for each of the electrical parameters to be tested, and briefly describe the testbench circuit setup and how it measures the specified parameter.

For an unaffiliated individual, testbenches can be build from USB instruments that provide analog and digital pattern generation and logic analyzer/oscilloscope functionality.

For unbuffered analog signals, an expensive active differential oscilloscope probe or a PCB with a high input impedance instrumentation amplifier will help.

As proof of concept, a single ended input is fed through the input single ended to differential buffer, through the fully differential transconductance stages and through the output Class AB buffer.

The $g_m C$ stages need to be configurable to operate both as a filter of the input signal and as quadrature oscillator that does not require input.

It must be possible to disconnect the input buffer from the internal stages, by connecting the inputs to V_{CM} instead.

Two of the shared analog pins are used to feed fully differential signals to and from the internal stages.

This will allow separate testing of the input and output buffer stages.

Measuring unbuffered differential analog signals may require an external instrumentation amplifier or, if feasible, auxiliary class AB amplifiers on chip driving the shared analog lines.

Tests:

1. input buffer off, output buffer on, filter off, differential input enabled (output buffer standalone)
2. input buffer off, output buffer on, filter off (output buffer offset and noise)
3. input buffer on, filter off, output buffer off, differential output enabled (input buffer standalone, measure V_{CM})
4. input and output buffers on, filter off (output connects to input)
5. input and output buffers on, filter on, filter configured as g_{mC} filter
6. input buffer off, filter and output buffer on, filter configured as g_{mC} oscillator
7. input buffer on, filter and output buffer on, filter configured as driven g_{mC} oscillator
8. input and output buffers off, filter off, differential output shorted (shared analog pin)

- and wire parasitics)
9. input buffer and filter on, output buffer off, differential output enabled (characterize filter, measure V_{CM})

Connections required for standalone (breakout) implementation

Indicate how the circuit is to be connected for individual testing outside of the eventual SoC application, and where test points may need to be added to access internal states of the circuit. Note where pad capacitance, wirebond inductance, and wire resistance from pad to circuit may affect measurement, and how to mitigate. Note where the circuit may need to be placed as close as possible to a pad.

The following pins would have to be made available:

from infrastructure moved to dedicated pins

Pin name	Use	Value
vdd3v3	analog power	3.3V nominal
vdd1v2	digital power	1.2V nominal
vss3v3	analog ground	0V
vss1v2	digital ground	0V
vssio	I/O ground	0V
ibias	bias current	to be discussed, 100nA ballpark
igmc	g_mC control current	to be discussed
vbias	common mode voltage	to be discussed, if adjustable

analog pins, close to pad (and instrumentation amplifier close to the pad on the PCB side)

Pin name	Use	Value
vcm	common mode voltage	maybe switchable to internal V_{CM}
vdifffp	differential analog I/O	"internal" differential voltage (buffered?)
vdifffn	differential analog I/O	"internal" differential voltage (buffered?)

digital input pins (some dedicated input pins could be absorbed into the scan chain)

Pin name	Use	Value
ena	master enable	
reset	reset digital logic	reset scan chain (and other logic?)
outbuf_en	output pin buffer enable	
inbuf_en	input pin buffer enable	input of next internal stage to V_{CM} if off
filter_en	internal g_mC filter enable	inbuf connected to outbuf if disabled
vdifff_en[1:0]	differential analog I/O enable	off, in, out, out_offset
vcmselect[1:0]	V_{CM} select	analog pin, internal, infrastructure
scanclk	scan chain shift clock	
scanctrl[1:0]	scan chain control	idle, shift, write, readback
scanin	scan chain in	

digital output pins

Pin name	Use	Value
scanout	scan chain out	

Test plan for standalone (breakout) implementation

Describe how the standalone circuit can be measured on a lab bench to verify that the circuit meets performance requirements for each specified electrical parameter.

- Serial data generator for digital control,
- instrumentation amplifiers for unbuffered internal differential analog signals
- Provide the possibility to (capacitively? By transformer??) inject AC into the power lines and the analog I/O lines to characterize PSRR, output resistance and such.